

Paraffin Deposition From Crude Oils: Comparison of Laboratory Results With Field Data

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Summary

In this study, we evaluate the characteristics of four reservoir fluids sampled from two offshore fields in the Gulf of Mexico as well as their propensities towards paraffins deposition due to temperature and/or pressure variations. State-of-the-art techniques, namely a cross polarized microscopy and a laser based solids detection system, have been used to measure the onsets of paraffin crystallization temperatures, also known as wax appearance temperatures (WAT) and/or cloud points. The results are presented, discussed and compared to reported field data. Explanations for the observed differences and/or similarities between laboratory and field WAT are proposed.

Introduction

In recent years, paraffin (wax) deposition has created diverse production problems in many of the world's oil-producing regions. Often the problem is more severe for offshore fields where solutions to rectify wax precipitation can be costly. Due to the relative difficulty and expense associated with operating waxy reservoirs, especially offshore, it has become necessary to understand the natures of petroleum waxes and the fundamental variables that affect their depositions during production and/or processing.

Waxes are essentially mixtures of long-chain hydrocarbons (*n*-paraffins) with carbon chain lengths ranging from C15 to C75+.¹⁻³ They are crystalline in nature and tend to crystallize/precipitate from crude oils at and below their cloud points. Crystallization is the process whereby an ordered solid structure is produced from a disordered phase, such as a melt or dilute solution (e.g., crude oil). It usually involves two distinct stages, namely nucleation and growth which may be considered separately.^{4,5}

As the temperature of the liquid solution or melt is lowered to its WAT, the energy of molecular motion becomes increasingly hindered, and the randomly tangled molecules in the melt tend to move closer together and form clusters of adjacently aligned chains. The paraffin molecules continue to attach and detach to and from these ordered sites until the clusters reach a critical size and become stable. This process is termed nucleation and the clusters are called nuclei. These nuclei are only stable below the melting/dissolution temperature of the wax as they are disrupted by thermal motion above this temperature. Once the nuclei are formed and the temperature is kept at or below the WAT, additional molecules are laid down successively on the nucleation sites and become part of the growing lamellar structure. This mechanism is called the growth process.

Nucleation can be either homogeneous, meaning that the sample is pure and the nucleation sites are time dependent or heterogeneous, which implies that all nucleation sites are activated instantaneously. The latter type is the most common in

crude oils where impurities such as asphaltenes, formation fines, clay and corrosion products act as nucleating materials for wax crystals.²

Although wax appearance temperatures (WAT) and pour point temperatures are specific thermodynamic properties for waxy crude oils, their relative positions/boundaries within pressure-temperature (P-T) diagrams as well as the rate (and amount) of wax deposition and accumulation between them are dependent upon a number of factors which include

- oil composition,
- temperature/cooling rate,
- pressure,
- paraffin concentration,
- molecular mass of paraffin molecules,
- occurrence of nucleating materials such as asphaltenes, formation fines, and corrosion products,
- water-oil ratio, and
- shear environment.

Recognizing that the onset of wax deposition is sensitive primarily to the waxy crude composition, it is important to review the properties and characteristics of the four major oil fractions namely, the paraffins, the naphthenes, the aromatics, and the polars (i.e., resins and/or asphaltenes). Examples of each of these classes of chemicals, their structures, and their boiling and melting points have been reported elsewhere.^{6,7} Briefly, both *n*-paraffins and *iso*-paraffins are flexible hydrocarbon molecules and, hence, tend to cluster together and precipitate from crude oil as wax solids. Being branched molecules, however, the *iso*-paraffins tend to delay the formation of wax nuclei and usually form unstable wax solids. Aromatics, on the other hand, are known to be good solvents for paraffinic waxes. Naphthenes, also known as cyclo-paraffins, are stiff and bulky in nature; they tend to disturb and/or disrupt the wax nucleation and growth processes. Unlike in the case of asphaltenes precipitation, resins do not have a direct effect on wax deposition. Finally, the presence of impurities and/or other amorphous solids (such as asphaltenes) in the oil usually induce a wax nucleation process^{2,8,9} as they tend to lower the energy barrier for forming the critical wax nucleus.^{4,5}

There have been reports,^{10,11} however, that the presence of asphaltene solids which are amorphous in nature help remediate paraffin deposition from crude oil, i.e., asphaltenes act as inhibitors. It appears that whether asphaltenes contribute to or prevent paraffin wax deposition depends on their relative concentrations, sizes and characters. Further investigations need to be undertaken to clarify this issue.

In the open literature,^{1,12-17} most of the laboratory measurements of the so-called cloud points are made on stock tank oil (STO) samples. These measurements are sensitive to the thermal history as well as the experimental technique used in the laboratory such as, the standard American Society for Testing and Materials (ASTM) D2500-88 or IP 219/82 methods, differential scanning calorimetry (DSC), cross polar microscopy (CPM), light transmittance (LT), viscometry, and cold finger (CF), to name a

TABLE 1–PARAFFIN STUDY SURVEY DATA

Sample Acquisition Data					Survey Data		
Field (Well)	Sample Type	Fluid Sample Depth (ft)	Pressure at Fluid Sample Depth (psi)	Temperature at Sample Depth (°F)	Survey Well Depth TVDF (ft)	Flowing Pressure at Survey Depth (psi)	Flowing Temperature at Survey Depth (°F)
High Island 557 (A-1)	Live oil	7263	2743	150	0	0	83
					60	843	87
					2000	1148	114
					4000	1592	139
High Island 557 (A-2)	Live oil	6427	2200	152	61	469	100
					3000	806	130
South Marsh Island 107 (A-6)	STO	9249	800	177	0	65	83
South Marsh Island 107 (A-10)	Live oil	6362	750	157	0	65	60
					4000	415	136

few. As such, the true definitions depend on the measurement principles of the equipment and techniques/procedures being employed.¹⁸

For instance, the ASTM methods rely on visual observation of the wax crystals; they require the liquids to be transparent in layers about 30 mm in thickness and, hence, often cannot be used to test dark crude oils (e.g., black oils). For black oils, WAT are usually inferred from viscometry, DSC, LT, or CPM. While the first three techniques require that sufficient paraffin crystallize before the properties of the crude oil change enough for the liquid/solid (L/S) phase transition to be detected, CPM allows for direct visual observation at the microscopic level (crystal size of the order of 1 μm)^{14,15} of the WAT. In other words, the CPM detects wax formation at the nucleation stage, whereas viscometry, DSC, and LT techniques ascertain the presence of paraffins more or less at the subsequent (early) stage of crystal growth. It is also possible that for low wax content crude oils, WAT and/or the presence of paraffin crystals may go completely undetected by viscometry, DSC and/or LT, leaving the investigator unaware of potential waxing problems.^{15,17}

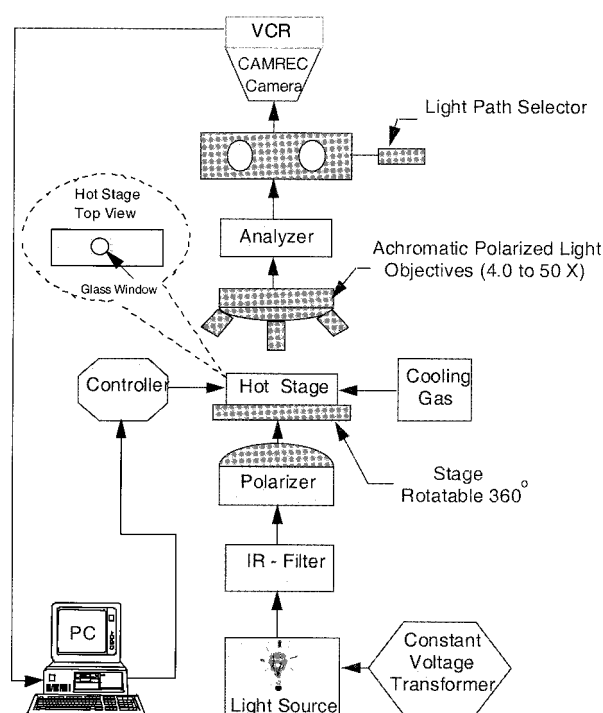
The relevance of laboratory measured WAT to actual field observations was not addressed until recently by Erickson *et al.*¹⁵ These authors used DSC, viscometry and CPM techniques to measure WAT for a variety of crudes (a total of ten dead oils sampled from different wells). They concluded that in almost all cases, the WAT measured by CPM were higher than those obtained by DSC and viscometry methods. More importantly, the WAT measured by CPM were in fair agreement with the field measured temperatures at which the wax deposition first begins in the corresponding wells (within 6°F). However, no explanations were provided for the observed match between quiescent WAT measured by CPM for the various STO samples and field wax deposition temperatures (FWDT) of the corresponding live fluids under flow conditions.

In this work, we report WAT, STO characterization, and compositional data of four reservoir fluids sampled from two offshore fields in the Gulf of Mexico. Like most of the Gulf of Mexico oils tested to date, these fluids are waxy, quite opaque and exhibit low paraffin deposition rates;¹⁷ thereby, sophisticated techniques such as CPM and laser based solids detection system (SDS) have been

used to determine their corresponding WAT. The as-measured WAT are compared to surveyed FWDT and explanations are proposed for the observed similarities and/or differences in this study.

Fluid Preparation and Compositional Analysis

A total of three live saturated fluids and one dead oil STO sample were supplied by Amerada Hess for the purpose of this study. These are identified in **Table 1** along with their corresponding field data. The high pressure sample cylinders for wells A-1, A-2, and A-10 were placed in an air-bath oven where they were equilibrated at 160°F and 5000 psig under intermittent rocking for 24 hours. Subsequently, a sample of each of the equilibrated fluids


Fig. 1–Schematic of the cross polarized microscope (CPM).

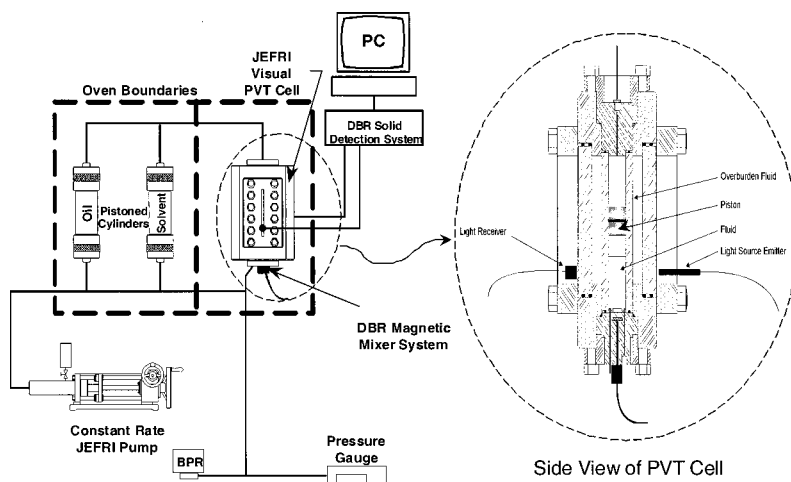


Fig. 2—Schematic of the solids detection system (SDS).

was analyzed for C_{35+} composition using a flash procedure. In this technique, an accurately measured volume of fluid was isobarically displaced into a pycnometer where its mass and density were evaluated. The pycnometer was then connected to a gas/oil ratio (GOR) single stage flash apparatus where the fluid was flashed to ambient pressure and temperature conditions. Next, the evolved gas phase was circulated through the residual liquid for a period of time to achieve equilibrium between phases. Following circulation, the volume of equilibrium vapor and the mass of liquid remaining in the pycnometer were measured. The vapor phase was resolved to C_5 by natural gas chromatography while the vapor C_{5+} fraction and the residual liquids were analyzed to C_{35+} . From the measured composition and total mass of each phase, the composition of the original live fluid was calculated by mass balance.

Meanwhile, the STO container for well A-6 was heated to 140°F in a water bath and hand rocked prior to withdrawing a

sample for compositional analysis by liquid gas chromatography to C_{35+} . Samples of the three live fluids were also flashed at atmospheric conditions and the resulting dead oils along with the supplied STO were characterized further as described in the subsequent section.

Measurements of STO Properties

The molecular weights of the four dead oils were determined using the freezing point depression method. The American Petroleum Inst. gravity values ($^{\circ}$ API) were determined by measuring the densities of the STO samples at 1 atm and 60°F. A precalibrated Anton Paar densitometer was used for this test. The corresponding pour point temperatures were measured following the standard ASTM D97 procedure.

Asphaltene Content (*n*-Pentane Insoluble Material). The Universal Oil Products (UOP) 614-80 method was used for this test.

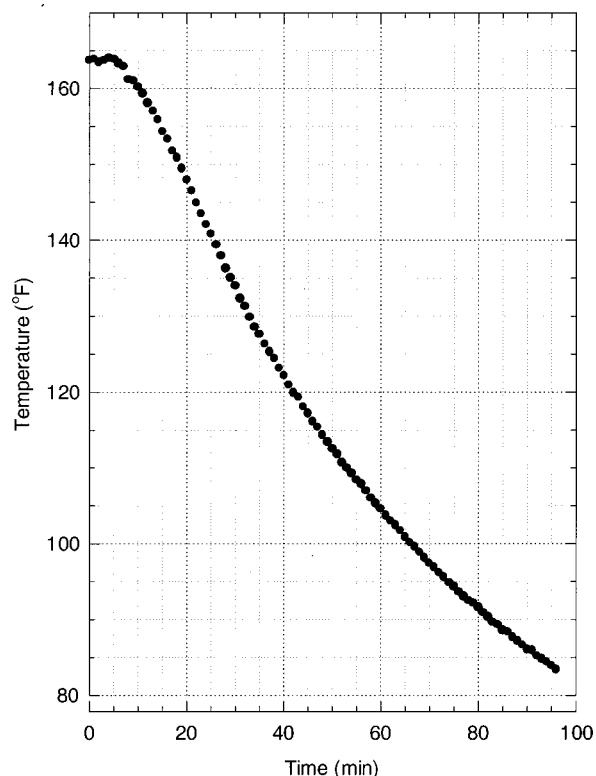


Fig. 3—Typical cooling curve for the solids detection system.

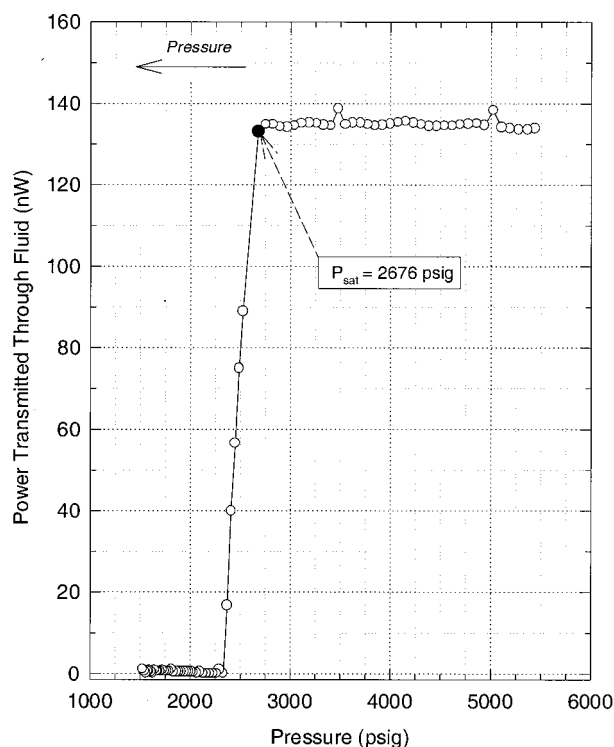


Fig. 4—Bubble point of HI 557/well A-1 fluid measured using SDS at 160°F (continuous depressurization mode).

TABLE 2—C₃₀₊ COMPOSITIONS OF LIVE FLUIDS AT 160°F AND 5014 PSIA

Field Well Component	HI 557				SMI 107	
	A-1		A-2		A-10	
	wt %	mole %	wt %	mole %	wt %	mole %
CO ₂	0.051	0.142	0.031	0.115	0.050	0.158
N ₂	0.072	0.313	0.000	0.000	0.000	0.000
C ₁	5.791	43.899	2.886	29.376	5.006	43.390
C ₂	1.390	5.622	0.984	5.342	0.662	3.063
C ₃	0.890	2.456	0.720	2.666	0.632	1.993
<i>i</i> -C ₄	0.378	0.790	0.331	0.929	0.328	0.785
<i>n</i> -C ₄	0.555	1.162	0.478	1.343	0.444	1.062
<i>i</i> -C ₅	0.453	0.763	0.424	0.960	0.385	0.741
<i>n</i> -C ₅	0.332	0.560	0.309	0.700	0.328	0.632
C ₆	0.769	1.085	0.748	1.416	0.704	1.135
McyC-C ₅	0.199	0.287	0.170	0.330	0.170	0.281
Benzene	0.376	0.586	0.491	1.027	0.279	0.498
Cycl-C ₆	0.206	0.306	0.173	0.343	0.161	0.272
C ₇	0.957	1.162	0.878	1.431	0.889	1.234
Mcycl-C ₆	0.615	0.762	0.504	0.838	0.475	0.673
Toluene	0.750	0.991	0.783	1.387	0.557	0.841
C ₈	1.459	1.554	1.249	1.785	1.355	1.649
C ₂ -Benzene	0.085	0.097	0.077	0.118	0.088	0.115
M&P-xylene	0.791	0.906	0.750	1.153	0.562	0.736
O-xylene	0.374	0.429	0.343	0.527	0.323	0.423
C ₉	1.471	1.394	1.177	1.499	1.390	1.507
C ₁₀	2.699	2.450	2.181	2.506	2.476	2.569
C ₁₁	2.499	2.068	1.980	2.071	2.231	2.111
C ₁₂	2.617	1.977	2.156	2.069	2.340	2.021
C ₁₃	3.347	2.326	2.853	2.530	2.926	2.325
C ₁₄	3.558	2.278	3.098	2.553	3.025	2.214
C ₁₅	3.869	2.285	3.545	2.729	3.196	2.157
C ₁₆	3.595	1.970	3.497	2.525	2.813	1.762
C ₁₇	3.613	1.854	3.751	2.551	2.902	1.703
C ₁₈	3.757	1.821	3.996	2.567	2.872	1.591
C ₁₉	3.661	1.693	4.398	2.678	2.800	1.480
C ₂₀	3.410	1.508	3.794	2.195	2.623	1.326
C ₂₁	3.114	1.302	3.903	2.151	2.353	1.124
C ₂₂	2.923	1.165	3.753	1.976	2.328	1.061
C ₂₃	2.733	1.045	3.602	1.814	2.172	0.950
C ₂₄	2.490	0.915	3.362	1.623	2.013	0.846
C ₂₅	2.328	0.821	3.168	1.468	2.086	0.841
C ₂₆	2.090	0.708	2.934	1.308	2.519	0.976
C ₂₇	1.985	0.646	2.777	1.192	1.114	0.414
C ₂₈	1.956	0.613	2.690	1.114	2.057	0.737
C ₂₉	1.650	0.499	2.318	0.927	1.591	0.550
C ₃₀	1.475	0.431	1.901	0.735	1.737	0.581
C ₃₁	1.404	0.397	2.051	0.767	1.630	0.527
C ₃₂	1.225	0.336	1.504	0.545	1.583	0.496
C ₃₃	1.189	0.316	1.421	0.500	1.384	0.420
C ₃₄	0.971	0.250	1.341	0.458	1.392	0.410
C ₃₅₊	17.876	3.062	14.525	3.162	29.046	7.620
MW (g/mol)		121.6		163.3		139.0
Density(g/cm ³)		0.749		0.871		0.839
GOR (SCF/BBL)		566.1		285.5		468.0

This method involved dissolving 5 g of a given STO in 5 mL of chloroform in a glass flask. While swirling the mixture, 200 mL of *n*-pentane was added slowly. The resulting mixture was then agitated for 5 min and eventually stored in the dark for 2 hours to precipitate asphaltenes (with occasional shaking). Subsequently, the content of the flask was filtered and additional volume of *n*-pentane was used to rinse the flask and poured on the filter cake until the wash became colorless (approximately 50 mL). The precipitate was thereafter washed with methylene chloride to dissolve the asphaltenes. The washing was stopped when the filtrate appears colorless. Finally, the methylene chloride was evaporated and the dissolved asphaltenes were dried and cooled before weighing.

Standard UOP Wax Content. The UOP 46-64 method involves dissolving 2 g of a given STO in petroleum ether, and the solution is clarified using fuller's earth. The petroleum ether is evaporated and the clarified oil re-dissolved in an acetone-petroleum ether mixture. This solution is then chilled to 0°F and filtered through a cold filter funnel, the wax being collected on an asbestos mat in the funnel. The wax is subsequently washed from the mat into a weighed flask, using hot petroleum ether. Finally, the petroleum ether is evaporated and the precipitated wax weighed.

WAT by Cross Polar Microscopy (CPM). As shown schematically in Fig. 1, the CPM consists of a light source, an IR filter, a polarizer, a temperature controlled (programmable) hot stage, an

TABLE 3—C₃₀₊ COMPOSITIONS (IN WT %) OF THE DEAD OILS

Field	HI 557				SMI 107			
Well	A-1		A-2		A-6		A-10	
Component	<i>n</i> -Paraffin as Applicable	Undefined Carbon No.	<i>n</i> -Paraffin as Applicable	Undefined Carbon No.	<i>n</i> -Paraffin as Applicable	Undefined Carbon No.	<i>n</i> -Paraffin as Applicable	Undefined Carbon No.
C ₃	0.11	0.00	0.10	0.00	0.04	0.00	0.05	0.00
<i>i</i> -C ₄	0.10	0.00	0.10	0.00	0.07	0.00	0.06	0.00
<i>n</i> -C ₄	0.18	0.06	0.18	0.04	0.14	0.00	0.12	0.02
<i>i</i> -C ₅	0.28	0.00	0.28	0.00	0.24	0.00	0.20	0.00
<i>n</i> -C ₅	0.23	0.00	0.23	0.00	0.23	0.00	0.20	0.00
C ₆	0.30	0.50	0.28	0.48	0.33	0.46	0.29	0.41
Mcyc-C ₅	0.19	0.00	0.16	0.00	0.18	0.00	0.16	0.00
Benzene	0.37	0.00	0.48	0.00	0.28	0.00	0.27	0.00
Cycl-C ₆	0.21	0.00	0.17	0.00	0.18	0.00	0.15	0.00
C ₇	0.37	0.66	0.33	0.59	0.37	0.67	0.36	0.57
Mcycl-C ₆	0.65	0.00	0.52	0.00	0.58	0.00	0.49	0.00
Toluene	0.80	0.00	0.81	0.00	0.66	0.00	0.58	0.00
C ₈	0.42	1.18	0.35	0.97	0.36	1.31	0.39	1.07
C ₂ -Benzene	0.09	0.00	0.08	0.00	0.11	0.00	0.09	0.00
M&P-Xylene	0.86	0.00	0.79	0.00	0.69	0.00	0.60	0.00
O-Xylene	0.41	0.00	0.36	0.00	0.39	0.00	0.35	0.00
C ₉	0.49	1.13	0.39	0.86	0.37	1.36	0.43	1.07
C ₁₀	0.65	2.32	0.51	1.79	0.51	2.53	0.63	2.04
C ₁₁	0.70	2.06	0.56	1.54	0.54	2.28	0.58	1.83
C ₁₂	0.73	2.16	0.59	1.69	0.60	2.32	0.60	1.93
C ₁₃	0.86	2.83	0.72	2.30	0.84	2.89	0.67	2.49
C ₁₄	0.92	3.00	0.80	2.48	0.77	2.73	0.71	2.56
C ₁₅	1.05	3.22	0.98	2.77	0.83	2.81	0.80	2.65
C ₁₆	0.99	2.97	0.99	2.71	0.70	2.40	0.71	2.33
C ₁₇	1.04	2.94	1.24	2.73	0.70	2.36	0.74	2.40
C ₁₈	1.01	3.13	1.14	3.08	0.60	2.32	0.67	2.44
C ₁₉	1.11	2.92	1.60	3.05	0.65	2.11	0.74	2.28
C ₂₀	0.97	2.79	1.23	2.79	0.55	1.99	0.62	2.21
C ₂₁	0.85	2.58	1.15	2.98	0.44	1.79	0.51	2.03
C ₂₂	0.85	2.37	1.12	2.85	0.51	1.71	0.58	1.93
C ₂₃	0.76	2.25	1.08	2.73	0.46	1.59	0.48	1.86
C ₂₄	0.68	2.06	0.99	2.56	0.46	1.48	0.43	1.75
C ₂₅	0.62	1.95	0.88	2.47	0.43	1.43	0.59	1.67
C ₂₆	0.48	1.82	0.73	2.37	0.32	1.46	0.39	2.33
C ₂₇	0.39	1.79	0.62	2.31	0.31	1.47	0.33	0.87
C ₂₈	0.49	1.66	0.72	2.12	0.44	1.45	0.56	1.67
C ₂₉	0.48	1.34	0.76	1.69	0.47	1.22	0.39	1.33
C ₃₀	0.36	1.26	0.47	1.54	0.30	1.24	0.35	1.53
C ₃₁	0.32	1.23	0.61	1.55	0.27	1.29	0.31	1.45
C ₃₂	0.28	1.07	0.42	1.17	0.43	1.12	0.34	1.37
C ₃₃	0.41	0.90	0.48	1.02	0.33	1.01	0.33	1.16
C ₃₄	0.36	0.71	0.53	0.89	0.26	1.07	0.30	1.20
C ₃₅₊		19.70		15.36		32.17		31.38
MW (g/mol)		242		255		268		262

analyzer, a video camera, a VCR, a video monitor and a PC. The hot stage is mounted on the stage of the microscope and connected to a liquid nitrogen tank to control the rate of cooling of the STO sample. This setup allows for visual determination of the WAT (or cloud point) as well as providing a record of the crystal growth and morphology. Typically, crystal sizes on the order of 1 μ m are detectable at the WAT.

The use of CPM for the study of wax crystallization is based on the fact that all crystalline materials rotate the plane of polarization of transmitted polarized light. Hence, by crossing two prisms on opposite sides of the oil sample, all light is initially blocked and the entire field of view appears black. On cooling, the crystallizing material appears as bright spots against this black back-

ground. This technique usually provides the most conservative (or highest) value of the crude oil cloud point temperature.^{14–16,18}

Each dead oil sample was preheated in a water bath to 140°F for 30 min so as to melt the waxes and homogenize the sample. Next, the corresponding sample container was hand rocked prior to transferring one drop of the oil to a special glass slide and covered with a cover slip. The sample was viewed under polarized light so that only crystalline materials (wax crystals) were visible. Once the sample was transferred to the slide and the field of view was brought into focus, the temperature of the slide was dropped at a programmable rate of 1.8°F/min; the entire run was videotaped simultaneously. The corresponding WAT were identified off

TABLE 4—MEASURED PROPERTIES FOR DEAD OILS

Field Well	HI 557		SMI 107	
	A-1	A-2	A-6	A-10
Properties				
MW (g/mol)	242	255	268	262
°API	34.2	32.9	29.6	28.4
UOP asphaltene content (wt %)	0.24	0.17	0.44	0.36
UOP wax content (wt %)	7.1	8.2	11.7	9.9
ASTM pour point (°F)	32.0	59.0	<21.2	48.2
CPM WAT (°F)	122.4	114.8	125.8	121.3

line as the temperatures at which the first detectable bright spots (in the order of $1\ \mu\text{m}$ in diameter) emerged in the black field of view.

WAT by SDS. The schematic of the SDS, which is based on light transmittance, is shown in Fig. 2. The SDS apparatus consists of two divisions, namely the *source* components and the *detector* components. The source constituents included a power source, a fiber bundle encased in a protective steel sheath and a collimating lens that distributed the laser into a beam of exacting dimensions. The detection components consist of a collimator that focuses the transmitted beam onto a receiver fiber bundle (again sheathed in steel) and a light power meter.

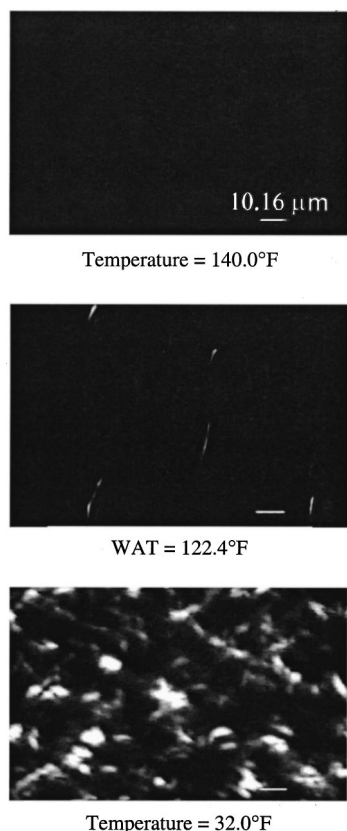


Fig. 5—CPM photomicrographs of HI 557/well A-1 STO at different temperatures (cooling rate=1.8°F/min).

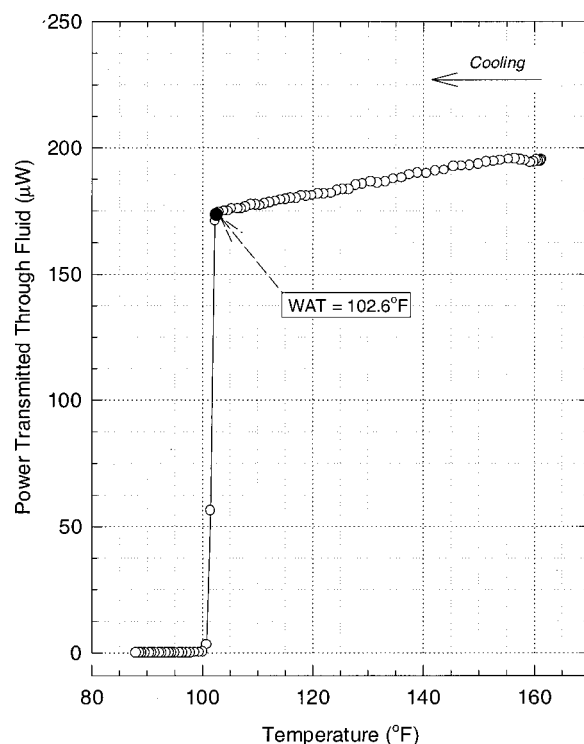


Fig. 6—Power of transmitted light through HI 557/well A-1 STO at 1 atm (cooling rate $\sim 1^\circ\text{F}/\text{min}$).

The source and detector components are mounted on a high temperature, high pressure visual pressure/volume/temperature (PVT) cell. The main body of the cell consists of a pistoned Pyrex glass cylinder 8 in. long with an internal diameter of 1.25 in. The effective working volume of the tube is approximately $130\ \text{cm}^3$. The glass cylinder is housed inside a steel shell with vertical tempered glass plates to permit visual observation of the internal

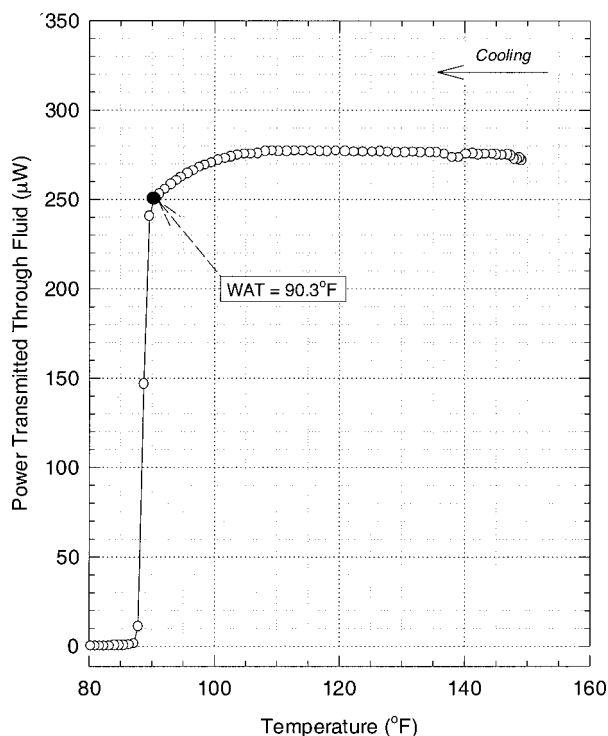


Fig. 7—Power of transmitted light through HI 557/well A-2 STO at 469 psia (cooling rate $\sim 1^\circ\text{F}/\text{min}$).

TABLE 5—WAT MEASURED BY SDS AT DIFFERENT PRESSURES

Field (Well)	Sample Type	Test Pressure (psia)	SDS WAT (°F)
	Live oil	1592	96
HI 557	Live oil	1148	97
(A-1)	Live oil	834	97
	STO	14.7	103
HI 557	Live oil	806	89
(A-2)	Live oil	469	90
SMI 107	STO	65	108
(A-6)			
SMI 107	Live oil	415	104
(A-10)	Live oil	65	109

Pyrex tube and its contents. The volume and hence the pressure of the fluid under investigation are controlled by a variable volume displacement pump. The same displacement fluid is connected to the outer steel shell to maintain a balanced differential pressure on the Pyrex cylinder. Equilibration of the fluid under investigation is achieved by means of a specially designed magnetically coupled impeller mixer. The entire PVT cell is mounted inside a temperature-controlled oven. The cell temperature is measured with a platinum resistance temperature detector (RTD). The temperature is read and controlled on a digital temperature indicator/controller and is accurate to within $\pm 0.2^\circ\text{F}$. Pressure is measured with a calibrated digital Heise pressure gauge.

A specially designed end cap mounted in the bottom of the cell allowed for the charging and sampling the reservoir fluid. The maximum operating pressure and temperature for the SDS are 15 000 psi and 360°F , respectively.

The system is controlled by a proprietary software package that accomplished two significant objectives. First, a computerized pump is used to maintain the system pressure and to inject precipitating solvents for asphaltene deposition studies. Second, the software (in real time) records and presents the system temperature, pressure, solvent volume, time, and, most importantly, the power level of the transmitted laser light from the detector.

The experimental procedure for measuring WAT by SDS involved charging a known volume ($\sim 70\text{ cm}^3$) of the test fluid at reservoir (or specified) temperature and pressure. The SDS components were then mounted across the windows of the PVT cell. Following this, the fluid was isothermally flashed and equilibrated

at the highest specified test pressure and the formed vapor phase was displaced from the PVT cell. The removal of the formed vapor phase from the cell was necessary to ensure constant liquid composition during the isobaric cooling experiment. Had the vapor phase remained in the PVT cell during the isobaric cooling, some of the vapor would condense as the temperature was lowered; thereby, the liquid composition would have changed continuously.

Following the removal of the vapor phase from the PVT cell, a reference scan was collected at reservoir temperature and the oil was allowed to cool isobarically under continuous mixing (mixer speed $\sim 1400\text{ rpm}$). The maximum cooling rate was in the order of $1^\circ\text{F}/\text{min}$ for this system (**Fig. 3**). Meanwhile, the average transmitted light power and the corresponding temperature were automatically recorded every minute using the computerized data acquisition system. The experiment was stopped a few degrees below the temperature at which a dramatic drop in the power of the transmitted light was observed.

Subsequently, the temperature of the cell was isobarically increased back to reservoir temperature and the power of the transmitted light (PTL) returned to the original baseline. This signified that paraffin deposition was reversible as expected. After equilibrating for 1 hour, the cell content was isothermally (at reservoir temperature) flashed to the next test pressure; the formed vapor phase was again removed from the cell and the isobaric cooling experiment was carried out as described above. This process was repeated for the various fluid samples at the specified pressures listed in Table 1. These pressures have been measured during field survey in each well.

The SDS can also be used to determine the bubble point pressure of the test fluid. By monitoring the PTL through the fluid during isothermal pressure drop (i.e., cell volume increase), the bubble point is determined as the pressure at which a sharp drop in transmittance occurs in a plot of PTL versus either cell pressure or volume (**Fig. 4**). Such a dramatic drop in PTL corresponds to the instantaneous and significant scattering of the laser light caused by emerging liquid-gas interfaces at and below the bubble point pressure.

Results and Discussion

Compositional Analyses. For ease of comparison, the C_{35+} compositions of the three saturated fluids are listed in **Table 2**. **Table 3** summarizes the compositions of the corresponding dead oils along with that of the South Marsh Island (SMI) A-6 STO. In this compositional work, information about *n*-paraffin as well as undefined carbon number component (such as *iso*-paraffins and

TABLE 6—COMPARISON OF LABORATORY QUIESCENT WAT TO FIELD DYNAMIC FWDT

Field (Well)	Live Oils			Dead Oils		
	Reported Depth of Wax Deposition, TVD (ft)	Pressure at Depth of Wax Deposition (psig)	FWDT (°F)	*SDS WAT (CWDT)** (°F)	CPM (OWCT)** (°F)	SDS (CWDT)** (°F)
HI 557 (A-1)	2534	1720	121	97	122.4	103
HI 557 (A-2)	730	1130	109	90	114.8	n/m**
SMI 107 (A-6)	3683	370	128	108	125.8	108
SMI 107 (A-10)	2399	580	118	106	121.3	n/m

*These values were interpolated based on the measured data at the specified test pressures.

**n/m : not measured; FWDT: field wax deposition temperature; CWDT: critical wax deposition temperature; OWCT: onset of wax crystallization temperature.

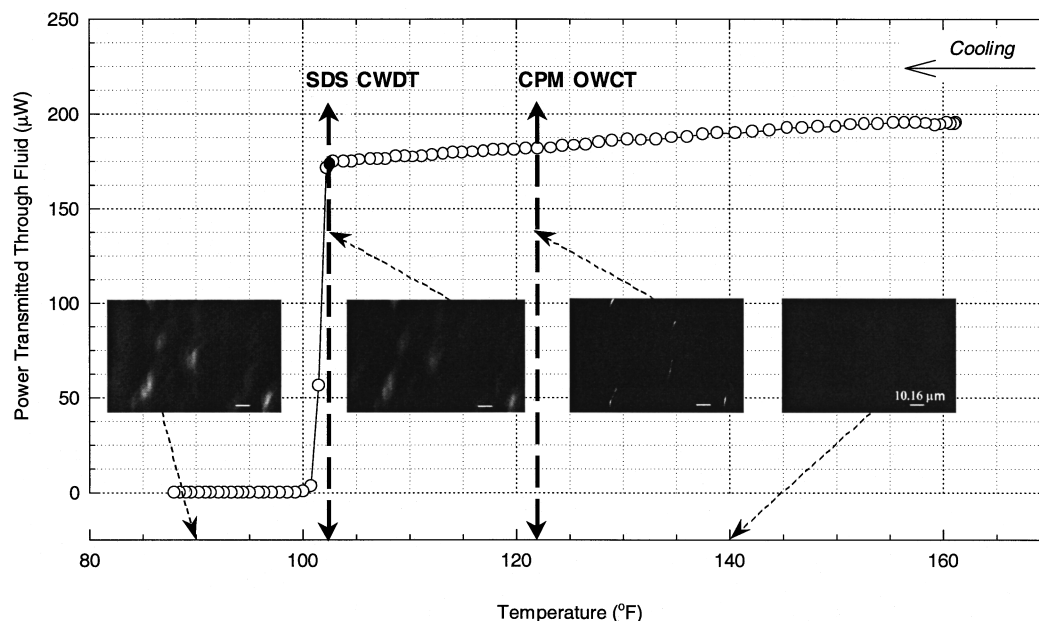


Fig. 8—SDS cooling trace and CPM photomicrographs at selected temperatures for HI 557/well A-1 STO.

naphthenes) concentrations in the dead oils is established from the measured chromatograms. The assignment of the weight percent data to the appropriate *n*-paraffin and/or undefined carbon numbers is based on the measured retention times of the calibration standard *n*-alkanes. The weight percent of an undefined carbon number corresponds to the sum of the area percent values of the peaks in between two consecutive *n*-alkanes. It is evident from Tables 2 and 3 that oils collected from the same reservoir but different wells have relatively different compositions. These differences may be attributed to geological factors and/or variances in temperature profiles and production histories for different wells.² The implications of the observed slight variations in oil compositions are interpreted below in relation to the measured STO properties.

STO Properties. Table 4 summarizes the measured molecular weights, °API, wax and asphaltene contents, and pour point temperatures of the four dead oils. Fig. 5 shows CPM photomicrographs of the High Island (HI) 557/well A-1 STO taken at three temperatures, namely 140, 122.4, and 32°F. As can be seen, paraffin crystals in the order of 1 μm in size emerge from the dark field of view as needles at 122.4°F. This temperature is termed the WAT for this oil. At 32°F, the number and average size of paraffin crystals grow significantly. Similar photomicrographs have been obtained for the other three dead oils; the corresponding WAT are summarized in Table 4.

As can be seen, the SMI 107/well A-6 dead oil has the highest molecular weight, asphaltene and wax contents and the largest C₃₅₊ fraction. Therefore, it is not surprising that this STO has the highest CPM WAT. The HI 557/well A-2 dead oil, on the other hand, has the lowest asphaltene content and the smallest C₃₅₊ fraction; thereby, its CPM WAT is observed to be the lowest.

The other HI 557/well A-1 and SMI 107/well A-10 dead oils have more or less the same CPM WAT. At first, this finding seems to be inconsistent with the fact that HI 557/well A-1 STO has a smaller C₃₅₊ fraction and lower asphaltene and wax contents than the SMI 107/well A-10 dead oil. However, a closer examination of the detailed compositional data presented in Table 3 reveals that the STO from HI 557/well A-1 has relatively more *n*-paraffins in the carbon number range C10 to C34 (i.e., active wax components) than that from the SMI 107/well A-10. Finally,

the SMI 107/well A-10 dead oil has lower asphaltene and wax contents and, in turn, lower CPM WAT than the SMI 107/well A-6 STO.

WAT for Dead and Live Fluids by SDS. During the process of flashing (isothermally at 160°F) the equilibrated HI 557/well A-1 fluid at the first test pressure of 1592 psia, the bubble point was measured to be 2690 psia (Fig. 4). This value is in good agreement with the 2743 psia reported (Table 1) at the sampling depth for this saturated fluid.

Fig. 6 shows a typical plot of PTL versus temperature measured for HI 557/well A-1 STO at atmospheric pressure. As the fluid temperature is lowered below its cloud point, paraffins begin to precipitate and block the path of the laser being transmitted through the fluid. This is seen in Fig. 6 as a dramatic drop in PTL.

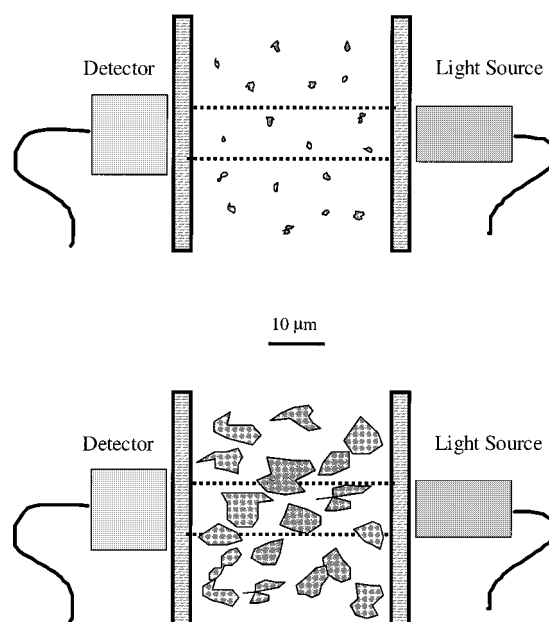


Fig. 9—Schematic illustration of the effect of wax crystal size on the transmitted light.

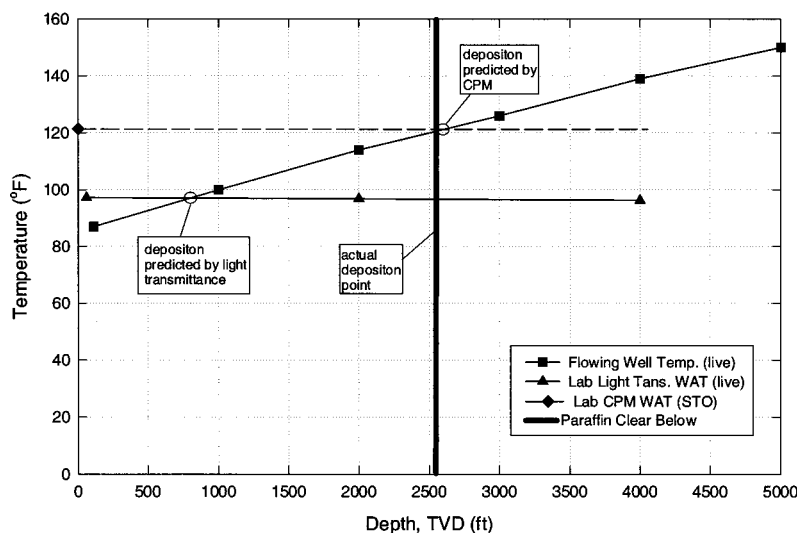


Fig. 10—Comparison of laboratory WAT results to surveyed/reported paraffin deposition data for HI 557/well A-1 fluid.

The initial gentle drop in PTL between 160 and 105°F is attributed to the expected slight increase of fluid density as the temperature is lowered. Fig. 7 shows a similar plot obtained for the HI 557/well A-2 fluid at 469 psia. The corresponding SDS WAT at the specified pressures for the different fluids are summarized in Table 5.

The SDS WAT are consistent with the trend typically observed and/or reported for waxy crude oils. It has been established that the presence of light ends in crude oils tend to depress the WAT.¹⁹ With decreasing test (flash) pressure, the concentration of light ends in the liquid phase decreases and, in turn, the WAT increases. However, because most of the light ends tend to flash at fairly low pressure (<100 psia) during differential liberation experiments, one would not expect to see strong variation in the cloud points above 100 psia as evidenced/confirmed by the measured WAT values for all three live fluids studied here.

Comparisons of Laboratory Measured WAT. Table 6 summarizes the quiescent WAT of the four reservoir fluid samples (both live and dead) determined by SDS and CPM as applicable. To this end, it is noted that the effect of mixer speed on the WAT has been investigated for these oils. The quiescent WAT by SDS (reported in Table 6) are generally 1 to 2°F higher than those obtained with the mixer turned on. These slight variations fall within the reproducibility limit of the SDS for WAT measurements.

For the STO samples (HI 557/well A-1 and SMI 107/well A-6), the CPM WAT are consistently higher than those measured using the SDS. The difference between the results from these two techniques is best illustrated in Fig. 8 where CPM photomicrographs taken at different temperatures for the STO from HI 557/well A-1 are superimposed on the corresponding light transmittance plot. While the CPM detects wax formation at the nucleation stage (i.e., when the paraffin crystals are about 1 μm in size), the SDS technique ascertains the presence of paraffins more or less at the subsequent (early) stage of crystal growth (i.e., when the wax crystals grow to about 10 μm in size as depicted in Fig. 9). Therefore, the WAT determined by SDS are termed critical wax deposition temperatures (CWDT); those detected by CPM are denoted as onsets of wax crystallization temperatures (OWCT).

Comparisons of Laboratory Measured WAT to Surveyed Field Data. The field wax deposition depths and, in turn, flow temperatures were reported by the lease operators on location during numerous well paraffin cuttings for each well. Flowing

pressure/temperature surveys with downhole electronic gauges provided the well temperatures at each maximum deposition point. The reported FWDT were surveyed from wells producing at relatively slow flow rates (essentially in the laminar regime). These results are reproduced in Table 6 for ease of comparison. Interestingly, the CPM OWCT are in good agreement with the wireline reported temperatures at the maximum depths where paraffin deposits were detected/confirmed (see Table 4 and Fig. 10). The surprising match between dead oil OWCT and live fluid FWDT is rather consistent with the findings of Erickson *et al.*¹⁵ as discussed earlier in the Introduction.

The impressive match between the CPM OWCT of the STO samples and the FWDT of the corresponding live oils is plausible for wells operating in laminar flow regimes. In fact, the general consensus is that shear environment can either delay or induce

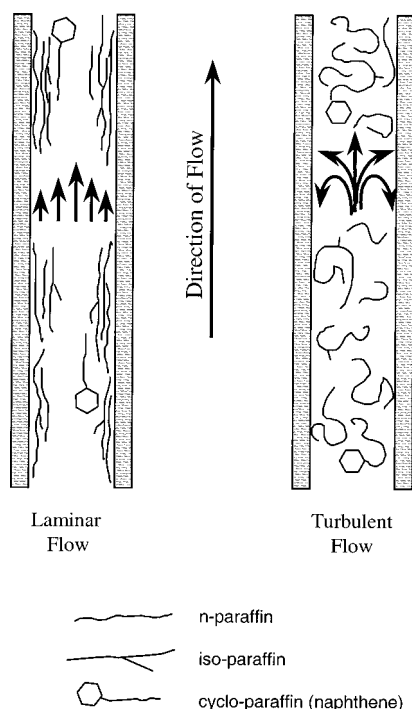


Fig. 11—Schematic illustrations of flow regimes in production wells and their effects on paraffin deposition.

wax deposition depending on the regime (i.e., turbulent versus laminar) of flow and/or agitation.^{20,21} In high shear flow environments (i.e., turbulent regime), the ability of wax constituents to congeal and reach a pour point or to plug piping is diminished as the nucleation process is disrupted. In low shear environment (i.e., laminar regime and/or closer to equilibrium/static condition), on the other hand, the flexible paraffinic molecules tend to be aligned adjacent to one another in the direction of flow, thereby, inducing/accelerating the nucleation/clustering process (Fig. 11). Hence, the severity of flow line plugging is usually observed to diminish for high shear environments. This effect can counteract and/or negate that of the light ends (for the live oils) which tends to depress the onset of wax crystallization temperature.

Conclusions

WAT were measured by CPM and SDS techniques for four fluids sampled from two offshore fields in the Gulf of Mexico. The WAT measured by CPM (also called OWCT) for the STO samples were found to be consistently and systematically higher than those obtained by SDS (better defined as CWDT). It was shown that while the CPM detects wax crystals at the nucleation stage, the SDS ascertains their existence at the subsequent early stage of crystal growth. The CPM WAT for the STO samples were found to match field wax deposition temperatures surveyed from the producing wells (i.e., under flowing live oil conditions, likely within and/or close to the laminar regime). It is speculated that laminar flow regime induces wax nucleation (crystallization) process and, in turn, counteracts the effects of light ends on the measured WAT.

Nomenclature

P_{sat} = saturation pressure, m/Lt², psi

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